



Validity and reliability of the Oura Ring Generation 3 (Gen3) with Oura sleep staging algorithm 2.0 (OSSA 2.0) when compared to multi-night ambulatory polysomnography: A validation study of 96 participants and 421,045 epochs

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ABSTRACT

Purpose: To evaluate the validity and the reliability of the Oura Ring Generation 3 (Gen3) with Oura Sleep Staging Algorithm 2.0 (OSSA 2.0) through multi-night polysomnography (PSG).

Participants and methods: Participants were 96 generally healthy Japanese men and women aged between 20 and 70 years contributing with 421,045 30-s epochs. Sleep scoring was performed according to American Academy of Sleep Medicine criteria. Each participant could contribute with a maximum of three polysomnography (PSG) nights. Within-participant means were created for each sleep measure and paired t-tests were used to compare equivalent measures obtained from the PSG and Oura Rings (non-dominant and dominant hand). Agreement between sleep measures were assessed using Bland-Altman plots. Interrater reliability for epoch accuracy was determined by prevalence-adjusted and bias-adjusted kappa (PABAK).

Results: The Oura Ring did not significantly differ from PSG for the measures time in bed, total sleep time, sleep onset latency, sleep period time, wake after sleep onset, time spent in light sleep, and time spent in deep sleep. Oura Rings worn on the non-dominant- and dominant-hand underestimated sleep efficiency by 1.1 %–1.5 % and time spent in REM sleep by 4.1–5.6 min. The Oura Ring had a sensitivity of 94.4 %–94.5 %, specificity of 73.0 %–74.6 %, a predictive value for sleep of 95.9 %–96.1 %, a predictive value for wake of 66.6 %–67.0 %, and accuracy of 91.7 %–91.8 %. PABAK was 0.83–0.84 and reliability was 94.8 %. Sleep staging accuracy ranged between 75.5 % (light sleep) and 90.6 % (REM sleep).

Conclusions: The Oura Ring Gen3 with OSSA 2.0 shows good agreement with PSG for global sleep measures and time spent in light and deep sleep.

1. Statement of Significance

Although many consumer wearables are being used in sleep research, the validity of the devices and their resulting sleep measures remains a source for concern. Indeed, many studies have investigated the validity of consumer sleep trackers, however, the characteristics of the study populations, including the small number of participants and 30-s epochs

used in analyses do not allow for generalizability of results. The present study investigates the validity of the Oura Ring Gen3 with the Oura Sleep Staging Algorithm compared to polysomnography in 96 participants with a total of 421,045 epochs. The results of the study are generalizable to the Japanese population and indicate the overall agreement of a smart ring with the gold standard of sleep assessment.

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2. Introduction

Sleep can be assessed subjectively using questionnaires or objectively through measurement of physiological sleep parameters. Objective sleep assessment is most commonly assessed using polysomnography (PSG), the gold standard of sleep evaluation [1]. PSG is a laboratory-based assessment, which allows for the recording of important physiological parameters including brain waves, eye movements, heart rate and rhythm, muscle movement, and oximetry [2]. The information from these measurements is combined to assess sleep stage, breathing and movements during sleep [2], which in turn can be used to diagnose disease or pathology.

Despite the importance of in-laboratory PSG in the field of sleep medicine, the method itself is not without limitations considering that it is expensive and that it must be conducted in a controlled laboratory setting which in turn may impact sleep habits [3]. For this reason, ambulatory PSG (home PSG) devices, which can assess sleep in a home setting, have been developed. Home PSG has, for obstructive sleep apnoea (OSA), shown diagnostic reliability and diagnostic precision comparable to in-laboratory PSG [4,5].

The last couple of years have seen the emergence of a new market of consumer wearable health- and sleep-trackers, including wrist-worn devices, rings, and headbands. Oura is a company that manufactures the Oura Ring, a smart ring equipped with numerous sensors that provide the user with information on global sleep parameters as well as sleep staging. In addition to these variables, the Oura Ring measures data on important physiological parameters, e.g., movement (step count), heart rate (HR), heart rate variability (HRV), respiration, and skin temperature, all of which are important in the context of sleep and its effect on normal physiology and health.

A number of validation studies have found good agreement between PSG and specific sleep measures, including time in bed (TIB) [6,7], total sleep time (TST) [6,8–12], wake after sleep onset (WASO) [6,13–15], sleep efficiency (SE) [7,8,10,14], and sleep onset latency (SOL) [6,10,14–16], obtained using consumer wearable health/sleep trackers. Despite these encouraging findings, only two validation studies [6,11] have found good agreement between PSG-derived sleep stages and corresponding sleep staging data obtained by consumer wearable devices. Although consumer wearable devices and the use of objective measures of sleep are becoming increasingly more important and relevant to highlight sleep patterns in large-scale studies [17–19], most

research that includes wearable device-derived sleep data focuses predominantly on global sleep measures. The consideration of sleep stages in large-scale public health research may be an important next step given associations between sleep architecture with age [20], as well as with insomnia [21], dementia [22], and diabetes [23].

Oura Sleep Staging Algorithm 2.0 (OSSA 2.0) includes accelerometry, HR, features of HRV, and temperature [24]. Multisensor wearables [25] add additional dimensions to sleep classification, including the discrimination of stages, and thus may allow for better agreement with PSG. The purpose of the present study was to 1) assess the agreement between multi-night ambulatory PSG and the Oura Ring Generation 3 (Gen3) with OSSA 2.0; 2) assess the reliability of the Oura Ring Gen3 with OSSA 2.0 by comparing rings worn on the non-dominant and dominant hand.

3. Method

3.1. Study overview

The primary objective of the study was to evaluate the performance of the Oura Ring Gen3 with OSSA 2.0 [24] to quantify sleep stages compared to a gold standard model (PSG with human sleep staging). The study was conducted between December 13, 2021 and February 7, 2022. Participants were asked to sleep in a designated hotel during their typical sleep schedule. For each night, participants were asked to wear three Oura Rings (one Generation 2 ring [Gen2m], and two Generation 3 rings (Gen3) worn on the left and right hand, respectively) in addition to the SOMNOtouch RESP PSG (SOMNOmedics GmbH; device described in detail below). Prior to bedtime, each participant completed a pre-sleep questionnaire detailing factors that could affect the participant's upcoming night of sleep. For each night when the participant was in bed and ready for sleep, they were asked to write down their bedtime (lights out) to mark the start of data collection. After wakeup in the morning, each participant was asked to write down their wakeup time (lights on) to indicate completion of data collection. Following completion of each night's data collection, participants removed the PSG sensors, completed a morning questionnaire, and sent the ring data via the mobile device(s).

The study aimed to include one hundred healthy adult participants between 20 and 70 years of age using targeted stratification according to age and sex as indicated in s Table 1.

Table 1
Participant inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
1. Japanese between 20 and 70 years old	1. Participants with known skin problems, such as allergic skin reactions, active eczema, or contagious skin conditions.
2. BMI between 18 and 30	2. Participants with known sleep disorders (including sleep apnea, periodic limb movement disorder, insomnia, or narcolepsy)
3. No evidence of sleep pathology a. STOP-Bang score (Q1-Q4) ≤ 2 at pre-screen and STOP-Bang score (Q1-Q8) ≤ 3 at visit 1 (V1)	3. Known allergies or previous reaction to titanium or medical-grade plastic
b. Epworth Sleepiness Scale <10	4. Current active illness (e.g. cold, flu, bronchitis, or suspected COVID-19)
c. GSDS average score <3	5. Implanted device (e.g. cardiac pacemaker, deep brain stimulator, etc.)
4. No diagnosed sleep disorder	6. Diagnosed psychoses (including Bipolar disorder and Schizophrenia)
5. Ability to read and understand the written consent form	7. Any other medical or psychiatric condition that PI deems exclusionary
6. Willing and able to cooperate with the study procedures for the duration of the study	8. Symptoms of sleep disordered breathing
7. Able to communicate and comprehend study instructions	9. Current pregnancy
8. Must have been using a smartphone for at least 6 months prior to the start of the study	10. Participant who routinely sleeps less than 5 h each night
9. Willing and able to apply PSG sensors prior to each night of sleep	11. Current shift or night work
10. Typical hours of sleep in the last month are between the hours of 8PM and 10AM	12. Has traveled through time zones in the past 4 weeks
	13. Is planning to travel through time zones during the duration of the study
	14. Is planning to be away from home for an extended period of time during the study period
	15. Participants who need to take sleep-inducing medication/sleep supplements (e.g., ornithine) during the study period
	16. Current use (within last month) of the following medications/classes: a. Sodium Oxybate
	b. Stimulants (Ritalin)
	c. Non-stimulant wake-promoting agents (e.g. modafinil, pitolisant, etc.)
	d. Antihistamines

Data selection was determined according to the successful collection of PSG measures; any available Oura Ring data without corresponding PSG data was thus excluded from analyses.

3.2. Participants

Participants were recruited based on the study's eligibility criteria (Table 1). Interested participants underwent an initial interview screening for pre-existing medical conditions and possible sleep disorders. Participants who passed the initial screening were invited for a first visit where they received additional information about the study and an explanation of their participation. All participants were informed that participation was entirely voluntary, and that participation could be discontinued at any time for any reason without any disadvantage or imposed penalty. All participants who understood the study, its risks, and agreed to participate were asked to provide written informed consent. Following informed consent, participants were checked for eligibility; all eligible participants were asked to confirm ring sizes. All participants were asked to participate in three consecutive nights of data collection.

Participants were, for each night of their participation, asked to maintain their typical sleep schedule as well as regular medication. Participants were further asked to refrain from strenuous exercise as well as caffeine/alcohol/tobacco use 6 h prior to a sleep study night.

Participants received a monetary compensation for their participation in the study and were reimbursed for transportation costs to/from the hotel. Additionally, participants were provided dinner before each night's PSG, and breakfast following each night's PSG. A participant that ended their participation before completion of all nights consented for researchers to use already collected data in any analyses. In the event of a participant's withdrawal from the study, pseudonymized participant data was deleted immediately following receipt of the participant's withdrawal form; such data is therefore not available for analysis purposes.

All personally identifiable information regarding participants was kept confidential and under strict access restrictions. Personally identifiable information was at no point transferred or revealed to the sponsor. All participants were assigned a unique, pseudonymized ID, which was attached to all data collected during the operation of the study after the Informed Consent had been signed. Non-identifiable PSG data and the results of a minimal subset of questions (i.e., gender, age, body mass index (BMI), hand dominance, ring size, and times of lights out/lights on) were shared through the University of Tokyo to Oura.

All research was performed in accordance with the relevant guidelines and regulations. The nature of the informed consent complies with the current version of the Declaration of Helsinki, the current requirements of Ethical Guidelines for Medical and Health Research Involving Human Subjects in Japan, whichever provided the greater protections to the participant. The study was approved by the Japan Conference of Clinical Research, Tokyo, Japan, on November 18, 2021 (certification no. 443). The study is also registered in the Japan Registry of Clinical Trials (JRCT) jRCT1030210486 (<https://jrct.niph.go.jp/en/latest-detail/jRCT1030210486>).

3.3. Polysomnography

Overnight PSGs were performed using the SOMNOtouch RESP PSG (SOMNOmedics GmbH) with the R&K headbox. The PSG montage included electroencephalography (EEG; F3-M2, C3-M2, O1-M2; 256 Hz sampling rate), bilateral electrooculography (EOG), and submental electromyography (EMG). Airflow, respiratory effort, and oxygen saturation were monitored using nasal cannula, dual thoracoabdominal belts, and continuous pulseoximetry, respectively. Participants were consented for three consecutive hotel nights of PSG data collection. A trained research nurse visited each participant's room in the evening for application of PSG sensors. Each research nurse was trained by a

certified sleep technician and a representative of Fukuda Denshi, the Japanese distributor of the SOMNOtouch RESP PSG prior to the start of the study. Electrode placement and PSG tracings were reviewed prior to bedtime to confirm the quality of data collection. In situations of inadequate PSG data collection or where the PSG device failed due to device issues unrelated to the participant, participants were not requested to complete the corresponding PSG night(s).

Sleep scoring (wake, N1, N2, N3, R) was done in 30-s epochs and performed according to criteria set out by the American Academy of Sleep Medicine (AASM) (2012) [26]. Sleep scoring was supported by the software Somnolyzer 24x7 [27]. PSG was scored by two expert analysts masked to Oura Ring recordings.

3.4. Oura Ring

The Oura Ring is a smart ring manufactured by Oura (Oura Health Oy, Oulu, Finland). The Oura Ring uses a 3D accelerometer, a negative temperature coefficient (NTC) thermistor, and infrared photoplethysmography (PPG) to measure body signals such as movement, body temperature, HR, HRV, and respiration [24]. These metrics are combined to yield information about sleep and wakefulness. All sensors are located on the inside of the ring on the palmar side of the finger. The device is powered by a USB charging base.

The Gen2m ring contains identical sensors and signals as the commercially available (at the time of conducting the study) product at www.ouraring.com. The Gen3 ring collects slightly different wavelengths of light than the Gen2m. Other than the aforementioned change, the exploratory versions used in the study were identical to the commercially available products. The hardware used in exploratory Oura Gen3 rings is identical to commercially available Oura Gen3 rings. Due to the nature of the study, research devices and/or associated hardware/software underwent sponsor-driven modifications. Device modifications included changing signal inputs or algorithm enhancements to enable continuous PPG data collection throughout the night; to extend battery life, standard settings turn off the PPG when registering large movements. Device modifications did not change the safety profile of the ring.

In this study, each participant wore a total of three rings; two rings were the exploratory Gen3 model (one on each hand) and the 3rd ring was the Gen2m randomized to either the dominant or non-dominant hand.

Each participant was equipped with (for the duration of their participation) three study-specific mobile devices pre-paired to each Oura Ring. Participants' personal mobile devices were at no point used in the study. The pre-paired mobile devices did not collect any protected health information or personally identifiable information about the participants. The designated mobile devices were used only to store and send data collected from their paired Oura Ring.

All participants' ring sizes were measured and confirmed at the time of the first visit.

The sleep measures considered for the present validation study were TIB, TST, SE, SOL, SPT, WASO, time spent in light sleep, time spent in deep sleep, and time spent in REM sleep. Oura provided researchers with summary sleep measures and 30-s epochs. Researchers did not have access to the raw data recorded by the accelerometer, NTC, or PPG sensors.

3.5. Epoch-by-epoch comparison

Thirty-second epoch-by-epoch (EBE) comparisons of sleep/wake states were used to calculate validity measures: sensitivity (proportion of PSG sleep epochs correctly identified as sleep by the Oura Ring), specificity (proportion of PSG wake epochs correctly identified as wake by the Oura Ring), predictive value for sleep [28] (PVS; proportion of true sleep epochs correctly identified by the Oura Ring as sleep), predictive value for wakefulness (PVW; proportion of true wake epochs

correctly identified by the Oura Ring as wake), and accuracy (proportion of PSG epochs that were correctly identified by the Oura Ring (i.e., true sleep and true wake). EBE data was obtained from the PSG and provided by Oura. The PSG epochs were coded as sleep if the respective epochs were scored as N1, N2, N3, or R whereas the Oura Ring epochs were coded as asleep if they recorded light (PSG equivalent: N1 + N2), deep (PSG equivalent: N3), or REM sleep (PSG equivalent: R). Epochs were coded as wake if the respective epochs were scored as W. Oura Ring provided epoch data in 30-s increments counting from every whole minute. Epoch data from the PSG was recorded in 30-s increments starting at the exact time of “lights-out” until the participant recorded “lights on” the following morning. Consequently, PSG epochs were rounded to the nearest half or whole minute mark (i.e., no epoch deviated more than 15 s from their original time stamp). EBE comparisons were restricted to the interval between evening lights out and morning lights on. Overall, 421,045 epochs (207,498 epochs for the Oura Rings worn on the non-dominant hand and 213,547 epochs for the Oura Rings worn on the dominant hand) were used in the 30-s EBE comparisons after removing 2534 epochs that could not be scored by the PSG.

Inter-device reliability was calculated as the epoch-by-epoch accuracy when comparing the Oura Rings worn on the non-dominant hand with the Oura Rings worn on the dominant hand. A total of 200,924 epochs were used for this purpose, excluding 19,197 epochs that could not be matched between the non-dominant and dominant sides’ rings.

3.6. Questionnaires and measures

Questionnaires (e.g., sleep disturbance, medical history, medication use, lifestyle habits, habitual sleep schedule) and anthropometric measures (e.g., height, weight, body fat percentage, abdominal circumference) relevant for the assessment of inclusion and exclusion criteria were administered at the first visit. The principal investigator checked participants’ questionnaire responses and measures against inclusion/exclusion criteria to assess eligibility. All questionnaires were administered electronically using a personal computer and all measures were taken by a research nurse.

In addition to the questionnaires at the first visit, eligible participants who had consented to participate in the study were asked to complete a pre-sleep questionnaire and a post-sleep questionnaire in conjunction with each study night. Anthropometric measures were repeated each morning following a study night.

3.7. Pre-sleep questionnaire

Prior to engaging in the PSG sleep night, each participant was asked to report on factors that could affect the participant’s upcoming night of sleep in a questionnaire that took approximately 5 min to complete. Such factors included the participant’s habits regarding caffeine, tobacco, medication, alcohol, and melatonin usage as well as the participant’s emotional state on the day of their study participation.

3.8. Post-sleep questionnaire

The morning following each PSG, the participant was asked to reflect on their night’s sleep using a post-sleep questionnaire. The questionnaire took approximately 5 min to complete.

3.9. Statistical analysis

Bland-Altman plots [29] were used to assess measurement agreement between equivalent measures (i.e., TIB, TST, SE, SOL, SPT, WASO, minutes spent in light sleep (PSG: N1+N2), deep sleep (PSG: N3), and REM (PSG: R) obtained using the PSG and the Oura Ring, respectively.

Each participant could contribute with a maximum of three PSG nights. Due to the repeated-measures nature of the study, within-participant means were created for each of the relevant sleep variables

(e.g., [TST(night 1) + TST(night 2) + TST(night 3)]/3). If a participant only had one PSG night, the sleep measures from that single night were used in analyses. A similar approach was used with the Oura Ring, where two within-participant means were calculated (one for the ring worn on the non-dominant hand and one for the ring worn on the dominant hand) for each sleep variable. This approach ensured that each participant was represented with one PSG measure and two Oura Ring measures for each sleep variable of interest. Of the 96 included participants (190 measures), 88 participants reported right-handed dominance, 6 participants reported left-handed dominance, and 2 participants reported ambidexterity. The final sample thus included 96 participants; 94 measures were available for analyses on the non-dominant hand and 96 measures for analyses on the dominant hand.

Results for all outcome measures were visualized in Bland-Altman plots with the difference (Y-axis: [PSG – Oura Ring]) against the mean (X-axis: [PSG + Oura Ring]/2). The lack of agreement between the respective measures (i.e., bias), was calculated as the mean difference between the PSG and the Oura Ring. A negative bias indicates overestimation whereas a positive bias indicates underestimation by the Oura Ring compared to the PSG. Limits of agreement were calculated as the mean difference ± 1.96 standard deviations (SD).

A clinically relevant difference of 30 min for TST and WASO, and 5 % for SE was set a priori [30]. The proportion of participants deviating from these cut-offs were calculated for the non-dominant and dominant hand as: [no. of deviating measures/total number of measures].

Paired t-tests were used to compare PSG and Oura Ring sleep measures. The prevalence-adjusted and bias-adjusted kappa (PABAK) which takes into account the bias index and prevalence index, respectively [31], was calculated to determine interrater reliability for epoch accuracy. PABAK is recommended for data where the proportion of sleep epochs is high relative to the proportion of wake epochs [30]. Interpretation of kappa values were as follows: 0–0.20 slight agreement, 0.21–0.40 fair agreement, 0.41–0.60 moderate agreement, 0.61–0.80 substantial agreement, and 0.81–1.00 almost perfect agreement [32].

Epoch-by-epoch concordance for sleep staging was calculated as the accuracy ([true positive + true negative]/total epochs).

Macro F1 score was calculated as the sum of F1 scores divided by the number of classes ($k = 4$), where the F1 score was calculated as: (true positive/[true positive + 0.5x(false positive + false negative)]).

All statistical analyses were performed using Stata/MP version 18.0 (StataCorp LLC, College Station, TX, USA). P values were two-tailed and considered significant if $p < 0.05$. Macro F1 score was calculated using Scikit-Learn.

3.10 Role of the funding source

The sponsor of the study (Oura) had no role in the analysis, and interpretation of data; in the writing of the report; or in the decision to submit the paper for publication. Oura was consulted only for the accuracy of technical details and specifications related to the Oura Ring.

4. Results

Participant characteristics are shown in Table 2. The mean ($\pm$ standard deviation) age and BMI of the study population were 41.9 (± 13.8) years and 22.8 (± 2.6) kg/m², respectively. The study population was equally represented among the two sexes (49 % men and 51 % women). The majority of participants had a university degree or higher (59.4 %), were right-handed (91.7 %), never-smokers (79.2 %), consumed caffeinated beverages on a regular basis (86.5 %), and were never drinkers or consumed alcohol no more than once per week (52.3 %). Self-reported habitual sleep duration was 7.1 (± 0.9) hours on workdays and 7.8 (± 1.0) hours on holidays (see Fig. 1).

Table 2
Participant characteristics.

Variable	
Number of participants	96
Age [mean (years ± s.d.)]	41.9 ± 13.8
Men (%)	49.0
BMI [mean (kg/m² ± s.d.)]	22.8 ± 2.6
Education (%)	
High school	17.7
Two-year college/technical college	22.9
University/Graduate school	59.4
Hand dominance (%)	
Right hand	91.7
Left hand	6.3
Ambidextrous	2.1
Smoking status (%)	
Never	79.2
Past	16.7
Current <20 cigarettes/day	4.2
Alcohol consumption (%)	
None	15.8
Less than once a month	9.4
2–4 times per month	27.1
2–3 times per week	18.8
More than 4 times per week	8.3
Caffeine consumption (%)	
Yes	86.5
No	13.5
Workday habitual sleep duration [mean (hours ± s.d.)]	7.1 ± 0.9
Holiday habitual sleep duration [mean (hours ± s.d.)]	7.8 ± 1.0

s.d. = standard deviation.

4.1. Comparison between the Oura Ring worn on the non-dominant and dominant hand, respectively, with polysomnography

The Oura Ring worn on the non-dominant or dominant hand did not significantly differ from polysomnography for the measures TIB, TST, SOL, SPT, WASO, time spent in light sleep, and time spent in deep sleep (Table 3). Oura Rings worn on both the non-dominant and dominant hand significantly underestimated the measures SE (non-dominant hand bias: 1.1 %; dominant hand bias: 1.5 %), and time spent in REM sleep

(non-dominant hand bias: 4.1 min; dominant hand bias: 5.6 min) (Table 4, Fig. 2, and Fig. 3).

The number (proportion) of measures that exceeded the 30-min ranges for the Oura Rings on the non-dominant hand were 11 (11.7 %) for TST and 13 (13.8 %) for WASO; the number of measures that exceeded the 5 % range for SE were 22 (32.4 %). The number (proportion) of measures that exceeded the 30-min ranges for the Oura Rings on the dominant hand were 14 (14.6 %) for TST and 15 (15.6 %) for WASO; the number of measures that exceeded the 5 % range for SE were 20 (29.0 %).

4.2. Epoch-by-epoch comparisons

Epoch-by-epoch comparisons were conducted using 207,498 and 213,547 epochs from Oura Rings worn on the non-dominant and dominant hand, respectively.

Oura Ring worn on the non-dominant hand had a sensitivity of 94.5 % (CI: 94.4 %–94.6 %), and a specificity of 73.0 % (CI: 72.5 %–73.5 %) (Table 5). The PVS was 95.9 % (CI: 95.8 %–95.9 %) and the PVW was 67.0 % (CI: 66.4–67.5). Accuracy was 91.7 % (CI: 91.6 %–91.8 %) with a Cohen’s kappa of 0.65. The bias index was 0.01 whereas the prevalence index was 0.72. The prevalence-adjusted and bias-adjusted kappa was 0.83.

Oura Ring worn on the dominant hand had a sensitivity of 94.4 % (CI: 94.3 %–94.5 %), and a specificity of 74.6 % (CI: 74.1 %–75.1 %). The PVS was 96.1 % (CI: 96.0 %–96.2 %) and the PVW was 66.6 % (CI: 66.0–67.1). Accuracy was 91.8 % (CI: 91.7 %–91.9 %) with a Cohen’s kappa of 0.66. The bias index was 0.02 whereas the prevalence index was 0.72. The prevalence-adjusted and bias-adjusted kappa was 0.84.

4.3. Sleep staging accuracy

Sleep staging accuracy of the Oura Ring worn on the non-dominant hand was 75.5 % (CI: 75.3 %–75.7 %) for light sleep, 88.6 % (CI: 88.5 %–88.8 %) for deep sleep and 90.6 % (CI: 90.5 %–90.7 %) for REM sleep (Table 5). Sleep staging accuracy for the Oura Ring worn on the dominant hand was 75.6 % (CI: 75.5 %–75.8 %) for light sleep, 88.9 %

Table 3
Sleep measures of the SOMNOtouch home polysomnography and the Oura Ring Generation 3 with Oura Sleep Staging Algorithm 2.0 worn on the non-dominant (n = 94) and dominant hand (n = 96), respectively, and their comparison using the paired *t*-test.

Non-dominant hand (n = 94)	SOMNOtouch PSG		Oura Ring Gen3		t	p value
	Mean ± SD	95 % CI	Mean ± SD	95 % CI		
Sleep measure						
TIB (min)	417.3 ± 32.8	410.5–424.0	417.6 ± 32.9	410.8–424.3	−0.41	0.6792
TST (min)	357.4 ± 46.7	347.8–366.9	355.3 ± 36.1	347.9–362.7	0.79	0.4311
SE (%)	86.3 ± 8.3	84.6–88.0	85.2 ± 6.1	83.9–86.4	2.25	0.0270
SOL (min)	23.1 ± 20.2	19.0–27.3	23.1 ± 16.1	19.8–26.4	0.03	0.9785
SPT ^a (min)	386.8 ± 31.2	380.4–393.2	388.9 ± 32.7	382.2–395.6	−1.74	0.0849
WASO ^b (min)	36.6 ± 37.8	27.5–45.8	34.8 ± 21.1	29.7–39.9	0.47	0.6367
Time spent in light sleep [N1+N2] (min)	208.4 ± 39.1	200.3–216.4	213.4 ± 31.2	207.0–219.8	−1.20	0.2338
Time spent in deep sleep [N3] (min)	70.4 ± 37.0	62.8–78.0	67.3 ± 22.0	62.8–71.8	0.88	0.3812
Time spent in REM sleep [R] (min)	78.6 ± 20.4	74.4–82.8	74.5 ± 19.1	70.6–78.4	2.22	0.0290
Dominant hand (n = 96)						
Sleep measure						
TIB (min)	417.5 ± 32.5	410.9–424.1	417.7 ± 33.0	411.0–424.4	−0.27	0.7907
TST (min)	357.9 ± 46.4	348.6–367.3	354.4 ± 37.8	346.7–362.0	1.41	0.1624
SE (%)	86.3 ± 8.2	84.7–88.0	84.9 ± 6.4	83.6–86.2	3.40	0.0010
SOL (min)	22.9 ± 20.0	18.9–27.0	22.0 ± 14.5	19.0–24.9	0.81	0.4199
SPT ^a (min)	387.4 ± 31.2	381.1–393.7	389.4 ± 33.4	382.7–396.2	−1.48	0.1425
WASO ^b (min)	36.3 ± 37.6	27.2–45.3	36.4 ± 25.3	30.3–42.5	−0.04	0.9687
Time spent in light sleep [N1+N2] (min)	208.8 ± 38.8	200.9–216.6	212.9 ± 28.5	207.2–218.7	−1.09	0.2770
Time spent in deep sleep [N3] (min)	70.0 ± 36.7	62.6–77.4	67.8 ± 21.5	63.4–72.1	0.65	0.5172
Time spent in REM sleep [R] (min)	79.2 ± 20.6	75.0–83.4	73.6 ± 19.1	69.8–77.5	3.10	0.0026

Abbreviations: CI=Confidence interval; REM = Rapid eye movement; SD=Standard deviation; SE=Sleep efficiency; SOL=Sleep onset latency; SPT=Sleep period time; TIB=Time in bed; TST = Total sleep time; WASO=Wake after sleep onset.

^a Compared “Total sleep period” (SOMNOtouch PSG) with “Sleep period time” (Oura Ring Gen3).

^b Wake after sleep onset was available for 69 participants.

Table 4

Agreement between equivalent SOMNOtouch PSG and Oura Ring Generation 3 with Oura Sleep Staging Algorithm 2.0 sleep measures with the corresponding Bland-Altman plot biases, standard deviations, 95 % confidence intervals of the biases, lower and upper agreement limits and the number of measures exceeding the agreement limits for the Oura Rings worn on the non-dominant (n = 94) and dominant hand (n = 96), respectively.

Non-dominant hand (n = 94)	Bias ± SD	Bias 95 % CI	Lower limit of agreement	Upper limit of agreement	N measures exceeding the limits of agreement
Sleep measure					
TIB (min)	−0.3 ± 6.8	−1.7–1.1	−13.6	13.0	5
TST (min)	2.1 ± 26.2	−3.2–7.5	−49.2	53.4	2
SE (%)	1.1 ± 4.8	0.1–2.1	−8.4	10.6	5
SOL (min)	0.02 ± 8.6	−1.7–1.8	−16.8	16.9	7
SPT ^a (min)	−2.0 ± 11.4	−4.4–0.3	−24.4	20.3	6
WASO ^b (min)	1.8 ± 30.7	−5.7–9.2	−58.4	62.0	4
Time spent in light sleep [N1+N2] (min)	−5.0 ± 40.5	−13.3–3.3	−84.3	74.3	6
Time spent in deep sleep [N3] (min)	3.1 ± 33.8	−3.9–10.0	−63.2	69.3	6
Time spent in REM sleep [R] (min)	4.1 ± 17.8	0.4–7.7	−30.8	39.0	5
Dominant hand (n = 96)					
Sleep measure					
TIB (min)	−0.2 ± 7.7	−1.8–1.4	−15.3	14.9	6
TST (min)	3.6 ± 25.0	−1.5–8.7	−45.4	52.6	2
SE (%)	1.5 ± 4.2	0.6–2.3	−6.8	9.7	8
SOL (min)	0.9 ± 11.4	−1.4–3.3	−21.4	23.3	5
SPT ^a (min)	−2.1 ± 13.6	−4.8–0.7	−28.8	24.7	6
WASO ^c (min)	−0.1 ± 28.3	−6.9–6.7	−55.6	55.4	6
Time spent in light sleep [N1+N2] (min)	−4.2 ± 37.5	−11.8–3.4	−77.7	69.4	4
Time spent in deep sleep [N3] (min)	2.2 ± 33.4	−4.6–9.0	−63.3	67.7	6
Time spent in REM sleep [R] (min)	5.6 ± 17.6	2.0–9.1	−29.0	40.1	7

Abbreviations: CI=Confidence interval; REM = Rapid eye movement; SD=Standard deviation; SE=Sleep efficiency; SOL=Sleep onset latency; SPT=Sleep period time; TIB=Time in bed; TST = Total sleep time; WASO=Wake after sleep onset.

^a Compared “Total sleep period” (SOMNOtouch PSG) with “Sleep period time” (Oura Ring Gen3).

^b Wake after sleep onset was available for 68 participants.

^c Wake after sleep onset was available for 69 participants.

(CI:88.7 %–89.0 %) for deep sleep and 90.6 % (CI:90.5 %–90.8 %) for REM sleep.

Fig. 4 shows hypnograms of sleep stages derived from the PSG and Oura Ring Gen3 for the three nights during which the Oura Ring had the highest (0.89), average (0.67), and the lowest (0.26) Macro F1 score.

4.4. Misclassification of stages

The Oura Ring worn on the non-dominant hand misclassified 17 % of PSG Wake as light sleep, 10 % of PSG N1+N2 as deep sleep, 35 % of PSG N3 as light sleep, and 23 % of PSG R as light sleep (Table 6). The Oura Ring worn on the dominant hand misclassified 16 % of PSG Wake as light sleep, 9 % of PSG N1+N2 as deep sleep, 35 % of PSG N3 as light sleep, and 24 % of PSG R as light sleep.

4.5. Inter-device reliability

The reliability of epoch-by-epoch comparisons between Oura Rings worn on the non-dominant and dominant hands was 94.8 % (CI: 94.7 %–94.9 %).

5. Discussion

The results of this validation study conducted over multiple nights in 96 participants show that a smart ring, the Oura Ring Gen3 with OSSA 2.0, has good agreement with PSG, the gold standard of sleep evaluation, for the key sleep measures TIB, TST, SOL, TSP, and WASO. Epoch-by-epoch comparisons show high accuracy, high sensitivity, and high PVS. Moreover, the Oura Ring Gen3 with OSSA 2.0 has good agreement for time spent in light sleep and time spent in deep sleep. Results were similar between Oura Rings worn on the non-dominant and dominant hand when compared to PSG, with a high (94.8 %) Oura Ring inter-device reliability.

5.1. Global sleep measures

Several validation studies have investigated the agreement between global sleep measures obtained from consumer sleep trackers or wearable devices and PSG. When looking at individual wearable device-measured global sleep parameters in isolation, results derived from these studies may indicate that consumer sleep trackers tend to show good agreement with PSG for TIB [6,7], TST [6,8–12], WASO [6,13–15], SE [7,8,10,14], and SOL [6,10,14–16]. However, most validation studies have shown significant differences between wearable-device derived sleep measures and PSG [6,7,11–16,33–37] with the understanding that consumer wearable devices tend to overestimate TST, and underestimate WASO. It is noteworthy that, to our knowledge, only two validation studies [6,15] in the same study have comprehensively investigated the agreement of all five of the global sleep parameters with PSG. The low number of comprehensive studies is surprising for two reasons: first, these study measures are readily available among several of the wearable devices that have already been validated. Second, and more importantly, comprehensive evaluations are crucial in order to allow for the correct interpretation of results under study-specific conditions.

The present study found no significant differences between Oura Ring-derived measures of TIB, TST, TSP, SOL, WASO, and SE with corresponding measures obtained from PSG. Detailed inspections of the resulting Bland-Altman plots of the rings of both the non-dominant- and dominant-hand reveal that the Oura Ring Gen3 with OSSA 2.0 tends to overestimate SE at lower proportions (<80 %), and underestimate both SOL and WASO at longer durations (>25 min and >30 min, respectively). This combination suggests that the Oura Ring Gen3 with OSSA 2.0 may not perform as well in those with poorer sleep quality (i.e., more time awake during the sleep episode, and longer time to initiate sleep). Patterns for TIB and TST do not show similar bias although the plots for TST reveal three observations (two for the non-dominant hand and one for the dominant hand) where the Oura Ring Gen3 with OSSA 2.0



Fig. 1. Flowchart detailing the inclusion and exclusion of study measures for the PSG and the Oura Ring Generation 3, respectively.

overestimates TST by more than 100 min. While it should be noted that all of these observations occur with $TST < 250$ min, they most likely represent individual outliers rather than a systematic bias.

Several validation studies have shown results related to TIB [6,7,12,15] or SOL [6,10,13–16,25,34,36,37]. Although many wearable devices significantly underestimate [6,13,36] or overestimate [34] SOL, some devices have good agreement with the gold standard [6,14,16]. In the present study, we found no significant difference between the Oura Ring Gen3 with OSSA 2.0 and PSG for either TIB or SOL. However, epochs were evaluated between self-reported “lights-out” and “lights on” times thus creating defined upper- and lower bounds. Successful measuring of TIB and SOL in naturalistic conditions may be challenging as it is inherently difficult for any wearable device to quantify the exact time a person intends to go to sleep (i.e., lights-out time) without external input. Indeed, no auto-detected lights-on/lights out times were provided by Oura, which in turn precludes the extrapolation of SOL-validity to naturalistic settings in which users rely on autodetection of sleep. SOL is a measure that can help identify conditions related to sleep or insomnia, and as such can be considered an important clinical parameter. It is therefore recommended that future studies evaluate SOL and TIB based solely on the automatic sleep tracking features of the wearable device in question to determine if indeed SOL and TIB can be quantified without participant input.

Sensitivity, specificity, PVS, PVW, accuracy and PABA were

calculated using all available epochs for the rings on the non-dominant (207,498 epochs) and dominant (213,547 epochs) hand, respectively. Two participants who were ambidextrous had their epochs from the Oura Rings worn on the left and right hand included in the analyses of the dominant hand. For sleep/wake detection, the Oura Ring Gen3 with OSSA 2.0 had a sensitivity of 94.5 % (non-dominant hand) and 94.4 % (dominant hand) and a specificity of 73.0 % (non-dominant hand) and 74.6 % (dominant hand), respectively. Specificity is high when compared to equivalent accelerometer-derived measures [38], whereas both sensitivity and specificity are high when compared with the results of other wearable-device validation studies [7,10–16,24,25,33–37,39–41] in which sensitivity ranges between 89.0 %–99.0 % and specificity between 8.1 %–85.0 %. Whereas other wearable devices may have high sensitivity but a low or very low specificity, there are few other studies that have shown a wearable device to perform well in both metrics. This performance is important to capture not only night-time awakenings with very little movement but also daytime naps, both of which may be commonplace among the elderly. It is noteworthy that the devices that in recent studies have exceeded both 90 % sensitivity and 70 % specificity are all rings [10,24,39]. The important question posed by de Zambotti et al. in a recent validation study of the Oura Ring [42], if “[...] the overall issue with specificity can be addressed or not by a multisensory approach in combination with sophisticated analytic methods”, thus seems to have an answer of “yes it can”. It remains to be

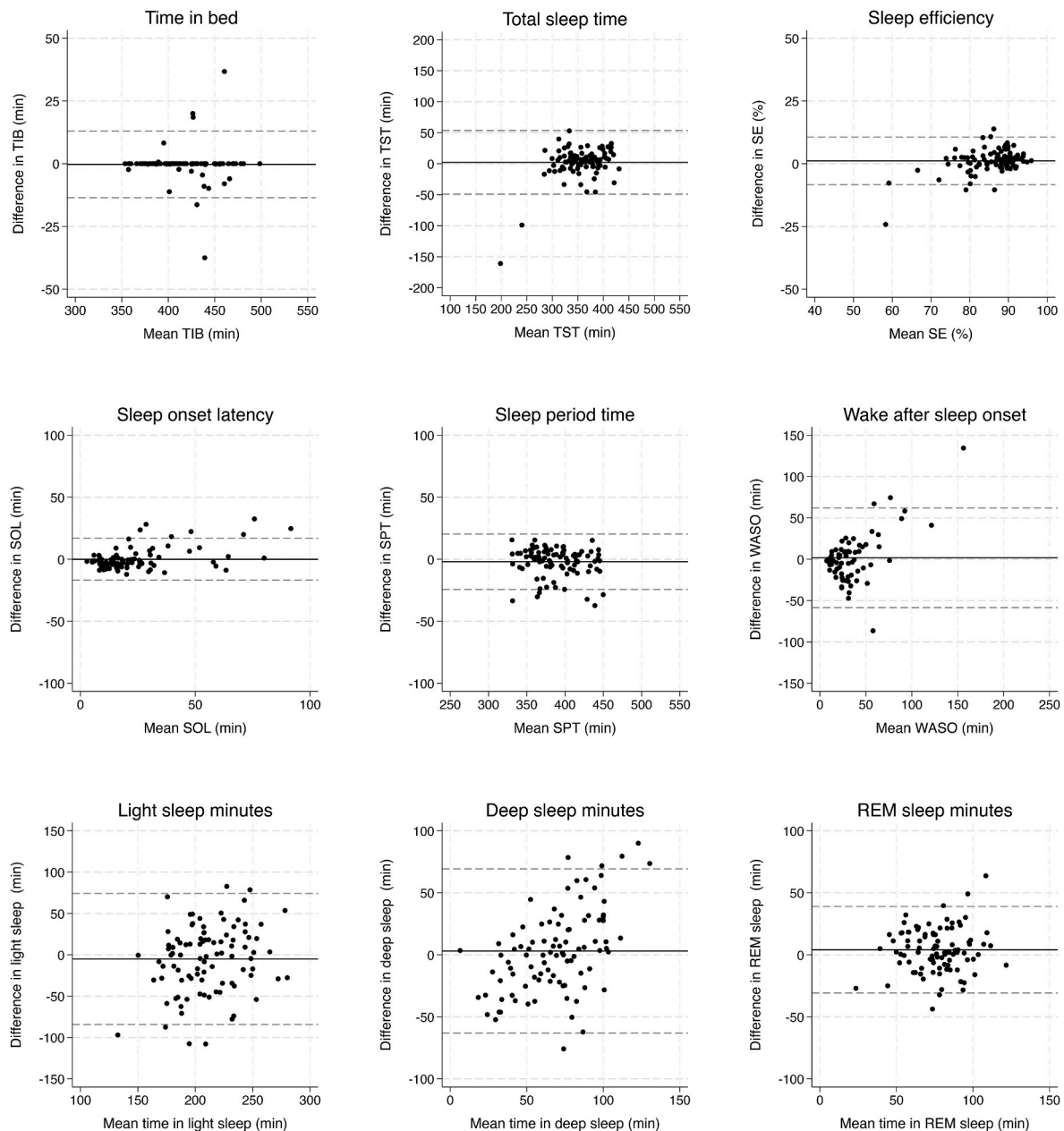


Fig. 2. Bland-Altman plots for time in bed (TIB), total sleep time (TST), sleep efficiency (SE), sleep onset latency (SOL), sleep period time (SPT), wake after sleep onset (WASO), minutes spent in light sleep, minutes spent in deep sleep, and minutes spent in REM for Oura Ring Generation 3 worn on the non-dominant hand. The Y-axis represents the difference between polysomnography (PSG) and the Oura Ring ($\text{PSG} - \text{Oura Ring}$) whereas the X-axis represents the mean value of the two devices ($(\text{PSG} + \text{Oura Ring})/2$). The solid line refers to the bias, and the dashed lines refer to the upper and lower limits of agreement calculated as the mean difference ± 1.96 standard deviations.

seen if additional improvements in sensitivity and specificity are related to improvement of existing sensors, incorporation of new sensors, or the fine-tuning of machine learning algorithms. Considering the advanced sensors already in place, it may well be that any improvement in sleep/wake detection will rely on the personalisation of machine learning algorithms, which have shown promising results in another smart ring-type wearable sleep tracker [39].

The accuracy of the Oura Ring Gen3 with OSSA 2.0 was 91.7 % (non-dominant hand) and 91.8 % (dominant hand). Other studies have reported accuracies for wearable devices between 87 %–92 % [7,10–16,24,25,33–35,37,39–41]. The found PABAK of 0.83 indicates near perfect agreement between the Oura Ring and PSG. Comparisons with other studies are, however, difficult owing to their reliance on Cohen's Kappa thereby omitting the important considerations of bias and data

distribution in which the number of epochs registering sleep outweighs those registering wake. Studies that have taken this into consideration report PABAK ranging between 0.75 and 0.80 [37].

5.2. Sleep stages

The sleep staging results obtained for the Oura Ring Gen3 with OSSA 2.0 were consistent for rings worn on the non-dominant and dominant hand, respectively. The Oura Ring Gen3 with OSSA 2.0 had good agreement with PSG for time spent in light sleep and time spent in deep sleep. Conversely, there was significant underestimation of the time spent in REM sleep. To our knowledge, only two validation studies have reported wearable devices that have good agreement with PSG for light, deep, and REM sleep [6,11], whereas most studies report significant bias

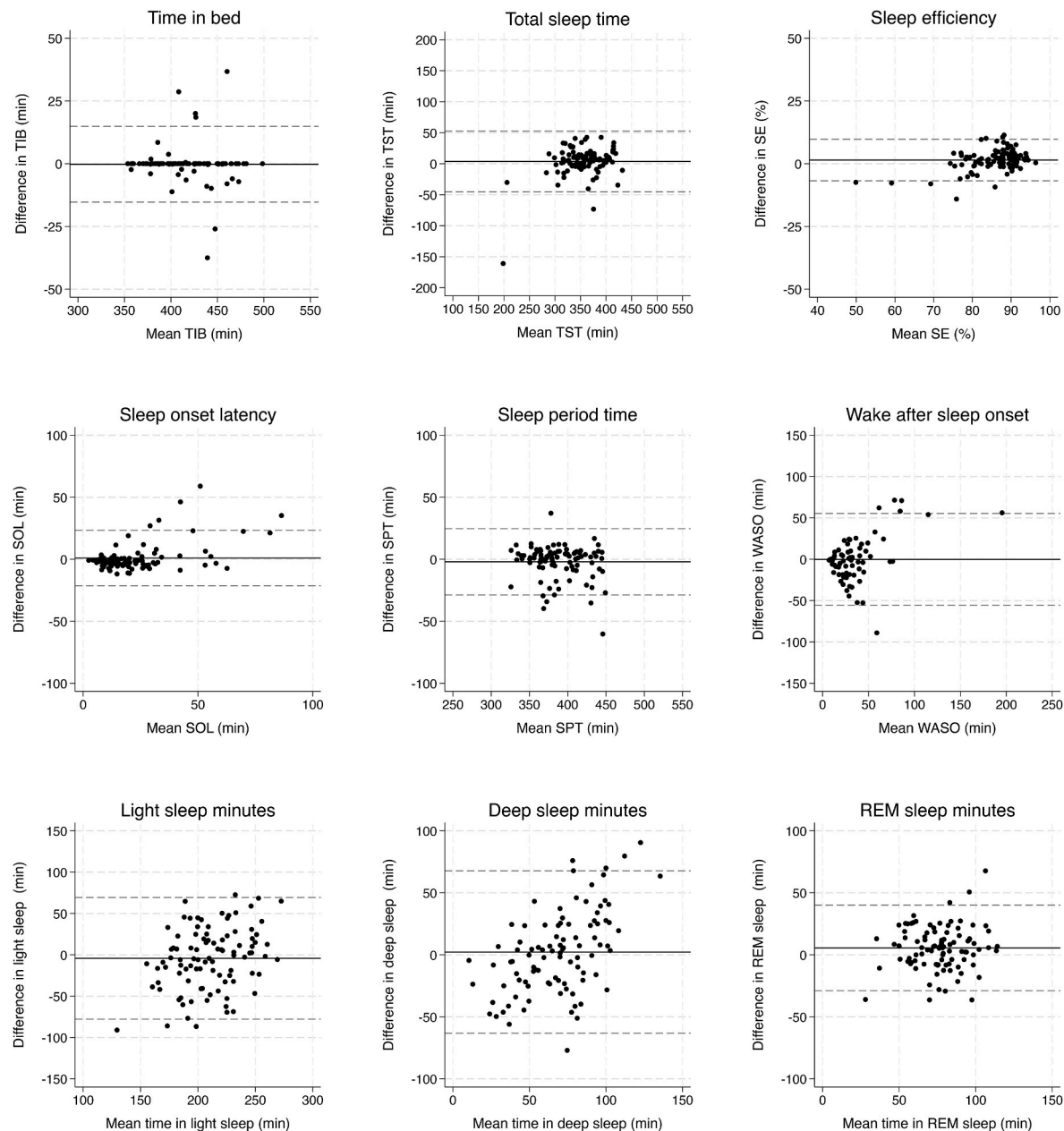


Fig. 3. Bland-Altman plots for time in bed (TIB), total sleep time (TST), sleep efficiency (SE), sleep onset latency (SOL), sleep period time (SPT), wake after sleep onset (WASO), minutes spent in light sleep, minutes spent in deep sleep, and minutes spent in REM for Oura Ring Generation 3 worn on the dominant hand. The Y-axis represents the difference between polysomnography (PSG) and the Oura Ring ($\text{PSG} - \text{Oura Ring}$) whereas the X-axis represents the mean value of the two devices ($(\text{PSG} + \text{Oura Ring})/2$). The solid line refers to the bias, and the dashed lines refer to the upper and lower limits of agreement calculated as the mean difference ± 1.96 standard deviations.

Table 5
Sensitivity, specificity, positive predictive values, negative predictive values, and accuracy of epoch-by-epoch sleep stage classification by the Oura Ring Generation 3 with Oura Sleep Staging Algorithm 2.0 worn on the non-dominant and dominant hand, respectively, compared to polysomnography.

Oura Ring Gen3 non-dominant hand	Sensitivity	Specificity	PVS	PVW	Accuracy	Prevalence- and bias adjusted Kappa
Sleep vs wake	94.5 %	73.0 %	95.9 %	67.0 %	91.7 %	0.83
Light sleep vs remaining epochs	76.7 %	74.4 %	75.4 %	75.7 %	75.5 %	0.51
Deep sleep vs remaining epochs	64.0 %	93.7 %	67.6 %	92.7 %	88.6 %	0.77
R sleep vs remaining epochs	72.6 %	94.9 %	77.2 %	93.6 %	90.6 %	0.81
Oura Ring Gen3 dominant hand						
Sleep vs wake	94.4 %	74.6 %	96.1 %	66.6 %	91.8 %	0.84
Light sleep vs remaining epochs	76.8 %	74.5 %	75.5 %	75.8 %	75.6 %	0.51
Deep sleep vs remaining epochs	64.5 %	93.8 %	68.1 %	92.8 %	88.9 %	0.78
R sleep vs remaining epochs	71.9 %	95.1 %	78.1 %	93.3 %	90.6 %	0.81

Abbreviations: PVS = predictive value for sleep; PVW = predictive value for wake.

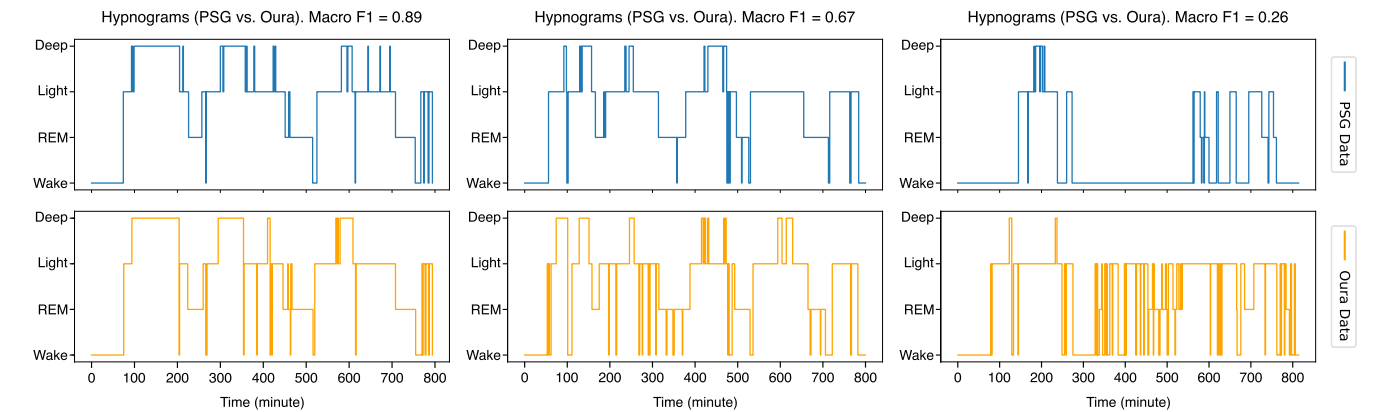


Fig. 4. Example hypnograms of 30-s epochs comparing polysomnography with the Oura Ring Generation 3 using Oura Sleep Staging Algorithm 2.0. The example hypnograms show sleep stages for three nights in which the Oura Ring Gen3 had the highest, average, and lowest precision as determined by the Macro F1 score.

Table 6
Misclassification of polysomnography sleep epochs by the Oura Ring Generation 3 with Oura Sleep Staging Algorithm 2.0 worn on the non-dominant and dominant hand, respectively.

Polysomnography	PSG Wake			PSG N1 + N2			PSG N3			PSG R		
	Light	Deep	Rem	Wake	Deep	Rem	Wake	Light	Rem	Wake	Light	Deep
Oura Ring Gen3 non-dominant hand	0.17	0.02	0.08	0.08	0.10	0.06	0.007	0.35	0.003	0.04	0.23	0.007
Oura Ring Gen3 dominant hand	0.16	0.01	0.08	0.08	0.09	0.06	0.008	0.35	0.003	0.04	0.24	0.007

Abbreviations: PSG = polysomnography.

for time spent in one or several sleep stages: light sleep [6,13,15,36,37], deep sleep [6,7,13,37], and/or REM sleep [7,9,37]. Significant underestimation of REM sleep by the Oura Ring Gen3 with OSSA 2.0 could potentially be explained by finger-twitches occurring during REM sleep [43]. Such occurrences may impact the accelerometer-based data during REM sleep and would be relevant for rings but not for wrist-worn devices. Indeed, another study using the Oura Ring also reported underestimation of REM sleep [40], whereas studies of wrist-worn devices have reported overestimation of REM sleep [7,9,40]. Conversely, it has been discussed that one difficulty of correctly assessing non-REM (NREM) sleep is due to movement which in turn affects the contact between the PPG sensor and the skin [44]. A ring would be far less susceptible to movement of sensors relative to the skin compared to a wrist-worn device. This highlights one very important advantage of the Oura Ring compared to wrist-worn devices in terms of sleep tracking and the reliance on multiple sensors.

Detailed inspection of the Bland-Altman plots is not clearcut but suggests that the Oura Ring Gen3 with OSSA 2.0 may overestimate light sleep when light sleep is shorter than 175 min and underestimate light sleep when it is longer than 240 min. Similarly, the Oura Ring Gen3 with OSSA 2.0 may overestimate deep sleep shorter than 40 min and underestimate deep sleep longer than 90 min.

Sensitivity for sleep stage detection was between 64.0 % (deep sleep measured on the non-dominant hand) and 76.8 % (light sleep measured on the dominant hand) whereas specificity ranged between 74.4 % (light sleep measured on the non-dominant) and 95.1 % (REM sleep measured on the dominant hand). It is thus noted that sensitivity is lower, and specificity is higher for sleep staging as compared to 2-stage (sleep/wake) discrimination. This is in line with a recent multi-device study that noted greater inconsistency in epoch sleep stage classification [37]; indeed, consumer devices showed an overall lower sensitivity and higher specificity for sleep stage detection compared to their sensitivity and specificity for only sleep/wake detection. Overall, the Oura Ring Gen3 with OSSA 2.0 misclassified 16–17 % of PSG Wake as light sleep, 9 %–10 % of PSG N1+N2 as deep sleep, 35 % of PSG N3 as light sleep, and 23 %–24 % of PSG R as light sleep.

The average precision of the Oura Ring Gen3 calculated using the

Macro F1 score was approximately 0.7. In the three example hypnograms, the lowest Macro F1 score was obtained for a night during which the participant had very little sleep and during which the Oura Ring predominantly misclassified wake as light sleep. Conversely, at average and at the highest Macro F1 score (indicating average to very high performance), the PSG and Oura Ring hypnograms align well which is in line with the results discussed above. It should be noted that the lowest Macro F1 was an outlier in the dataset and would represent only a fraction of nights from otherwise healthy participants.

5.3. Possible explanations for the good agreement between Oura Ring and polysomnography

Earlier validation studies to assess sleep and wake periods were done on wearables with single sensors (e.g., accelerometers only). However, advancement in technologies allowing wearables to take advantage of multiple sensors has ultimately led to improved algorithms of sleep detection, sleep/wake discrimination and sleep staging functionality. Indeed, a systematic review and meta-analysis showed that the sleep/wake performance of Fitbit devices improved when they were equipped with sleep staging functions [45]. Moreover, the inclusion of HRV in addition to actigraphy significantly improved Kappa and sensitivity to Wake compared to actigraphy alone [44]. Indeed, HRV decreases throughout NREM sleep and increases during REM sleep [46].

An important point that has been raised by Fonseca et al. [44] about the use of HRV in sleep assessment is the question of usefulness of this measure in individuals with medical conditions that distort the signal. One pre-requisite for a stable HRV measure is sinus rhythm [46]. Although participants in the present study were predominantly healthy, it remains to be investigated how well sensor-derived measures and their dependent algorithms perform in clinical populations with conditions that may impact HRV-measures.

One explanation for the Oura Ring’s overall good agreement with PSG is the reliance on multiple sensors to assess sleep and wake periods. A recent publication [24] revealed that Oura Ring’s latest algorithm utilises not only the accelerometer (for movement), but also NTC (for temperature), and PPG (for heart rate, HRV, and respiration) to assess

sleep and wake. The addition of temperature and HRV to a model relying on accelerometer data alone improved sensitivity and specificity in 2-stage sleep classification and improved accuracy, sensitivity, and specificity during 4-stage classification in Oura Ring's latest sleep staging algorithm [24].

The found agreement between Oura Ring Gen3 with OSSA 2.0 and PSG can thus be due to several factors, including the quality of sensors and/or the proprietary algorithm used. It should be noted that this cannot be tested as the authors did not have access to individual sensor data or the algorithms themselves.

5.4. Clinical relevance

The clinical relevance of wearable devices has been discussed in terms of pre-specified cut-offs (<30 min for TST and WASO, and <5 % for SE) that are based on the difference between measures obtained from PSG and the wearable device [30]. In the present study, the proportion of measures that exceeded these differences were 11.7 %–14.6 % (TST), 13.8 %–15.6 % (WASO), and 29.0 %–32.4 % (SE), respectively. With the exception of WASO (non-dominant hand), the standard deviations of the biases for TST (non-dominant and dominant hand), WASO (dominant hand), and SE (non-dominant and dominant hand) indicate that the inter-individual variability remains within the above suggested clinical limits. This would be relevant information also for measures (i.e., TST and WASO) that show good agreement between the Oura Ring Gen3 with OSSA 2.0 and PSG. Despite certain caveats [30] regarding the definition of these clinical limits, the present study's results remain encouraging for the Oura Ring's applicability in clinical research and lifestyle modification. Indeed, based on the results of the present study, it would appear promising to consider the Oura Ring in clinical trials focusing on sleep parameters. However, the question of using the Oura Ring in clinical practice would necessitate additional considerations: although clinical diagnosis and treatment based on wearable-device data would require regulatory approval, the present study may support the prospects of using the Oura Ring in the context of lifestyle improvement and disease prevention. One example could be to create a personal sleep record that can be shared with health practitioners and used as the basis for informed lifestyle modifications.

A decision that would undoubtedly improve any wearable device's clinical relevance would be for the proprietary algorithm to be made public and for the raw sensor data to be made accessible to researchers. This would allow researchers to not only create personalized algorithms depending on the characteristics of the study population, but to share these algorithms within the research community. Personalized algorithms have shown superior results compared to general algorithms [39] and could possibly improve even more.

5.5. Benefits of using wearable devices in future research

The use of wearable devices in research is entirely dependent on the purpose of the study requirements. There will be study designs that will accept nothing except for the detailed sleep measures that only PSG can provide. However, there is also a field of sleep research that is interested in broader questions related to individual- and population-level assessment of sleep parameters obtained over months or years, and their association with health outcomes. The effortlessness with which longitudinal wearable device-data can be collected across tens of thousands or even hundreds of thousands of individuals currently has no match among medical devices or research-grade accelerometers. Not only is sleep and lifestyle data synchronised with a smartphone application, but corresponding data, albeit limited or pre-processed using proprietary algorithms, can comparatively easily be downloaded by the researcher. Moreover, despite any concerns as to the general validity of wearable devices' sleep tracking abilities, there is little doubt that these are improving, as are the designs of the studies aiming to validate said devices.

Another important distinction is the possibility to use wearable devices for 24-h sleep assessment. Indeed, the high sensitivity and specificity indicates the possibility to detect also daytime naps. Although very attractive for understanding lifestyle patterns, the results of the present study do not allow for any conclusion of how the Oura Ring Gen3 with OSSA 2.0 performs at very short TSTs (e.g., 1–2 h). Additional studies investigating the validity of daytime napping are warranted. Another important point for future research is device reliability. Reliability is a key metric for any device aiming to be used in research. The present study shows a very high reliability of the Oura Ring Gen3 with OSSA 2.0 across 200,924 compared epochs between rings worn on the non-dominant and dominant hand, respectively. High reliability ensures that the device performs across multiple nights and participants, and irrespective of hand dominance.

5.6. Limitations

This study has some limitations. First, the results of this validation study may be difficult to generalize to naturalistic conditions. Although the study was not conducted in strict laboratory settings, certain elements of the protocol required preparation by experienced clinical research coordinators and imposed certain restrictions on participants (e.g., alcohol intake, physical activity) that would otherwise not occur. Second, PSG is not a suitable comparator for naturalistic conditions that may require weeks or months of data collection. Nevertheless, the results of the present validation study are valid in the context of multi-night PSG with the important consideration of sleep stages. Third, although unattended home PSG devices have been shown to differ from in-laboratory PSG on certain sleep parameters, i.e., longer total sleep time, higher sleep efficiency, decreased stage 1 sleep and an increase in REM sleep [47], the participants in the present study had all sensors applied by a trained research nurse and followed a standardized protocol thereby minimizing any bias. Fourth, the Oura Sleep Staging Algorithm 2.0 is proprietary and can therefore be changed at any time. Any updates to the algorithm that has been validated in this study thus warrant additional validation studies. However, it may be assumed that any updates to an existing algorithm is done to improve the algorithm's performance rather than to weaken it.

5.7. Strengths

This study also has several strengths. First, the study was conducted according to the latest guidelines on how to perform and interpret wearable device validation [48]. Second, participants were selected based on well-defined inclusion criteria. Third, they represent a wide age range and equal balance of the two sexes which allows for generalizability to a predominantly healthy Japanese population of men and women between the ages 20–70 years. Fourth, participants slept in similar conditions with standardised dinners and on-site confirmation that participants did not engage in activities that may have negatively influenced the investigation (e.g., excessive physical activity or pre-bedtime intake of alcohol and caffeine). Fifth, the study design allowed for confirmation of electrode and ring placement by a sleep technician/trained research nurses during all nights. Finally, it is one of the largest wearable-device validation studies to date with multi-night PSG of 96 participants and a total of 421,045 epochs.

6. Conclusions

The Oura Ring Gen3 with Oura Sleep Staging Algorithm 2.0 demonstrates good agreement with PSG for TIB, TST, TSP, WASO, time spent in light sleep and time spent in deep sleep. The Oura Ring Gen3 with OSSA 2.0 has high sensitivity and specificity with almost perfect agreement with PSG for 2-stage sleep classification, and an inter-device reliability of 94.8 %. The Oura Ring Gen3 with OSSA 2.0 is thus a very promising device to consider for studies in which global sleep

parameters are required, in particular, for data collection occurring over several nights and in which a moderate to almost perfect agreement with PSG for sleep staging is acceptable.

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Auhor contribution statement

TS and AKS were responsible for the conception and design of the study. TS analysed the data. KM, HNT, UIC and AKS interpreted the data and results. TS drafted the manuscript. KM, HNT, UIC and AKS critically revised the manuscript for important intellectual content. All authors approved the final version of the manuscript. TS is the guarantor of this work and, as such, had full access to all the data analysed in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. TS had final responsibility for the decision to submit for publication.

Data availability statement

The authors cannot publicly provide access to individual data due to participant privacy in accordance with ethical guidelines. Additionally, the written informed consent obtained from study participants does not include a provision for publicly sharing data. Qualifying researchers may apply to access a minimal dataset upon reasonable request by contacting the corresponding author at the following email address: thomas.svensson@med.lu.se.

CRedit authorship contribution statement

Thomas Svensson: Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kaushalya Madhawa:** Writing – review & editing, Visualization. **Hoang Nt:** Writing – review & editing, Visualization. **Ung-il Chung:** Writing – review & editing, Supervision, Software, Resources. **Akiko Kishi Svensson:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

Thomas Svensson reports financial support was provided by Oura Health Oy. Ung-il Chung reports financial support was provided by Oura Health Oy. Akiko Kishi Svensson reports financial support was provided by Oura Health Oy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.sleep.2024.01.020>.

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